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RESEARCH ARTICLE

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Co-design as a participatory approach for enhancing HPV health literacy among youths: a program evaluation

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ABSTRACT

Background: Despite decades of vaccine availability and the potential consequences of infection, human papillomavirus (HPV) vaccination among youth in the United States remains below herd immunity threshold. In response to suboptimal HPV vaccination rates the Vaccine Literacy Campaign (VLC) implemented a five-week series of HPV vaccination education co-design workshops to engage youth in creating peer vaccine communication. The co-design cohort included thirteen 16 to18-year-olds recruited from a community organization. The goals of the pilot were two-fold: 1) engage participants in co-design activities to co-create resource that promote health literacy, and 2) increase youth confidence and health literacy in relation to HPV vaccination.

Methods: Each co-design session was evaluated with a mixed-methods questionnaire to collect participant feedback on implementation and co-design content. Evaluation measured changes in confidence on HPV vaccine knowledge and satisfaction of co-design sessions. Qualitative evaluations focused on best practices for youth engagement strategies and understanding the root of youth vaccine hesitancy and perceived vaccine myths. Each session included co-collaboration amongst participants to develop an educational product promoting uptake of the HPV vaccine.

Results: Participant knowledge and confidence scales on HPV vaccination increased during the five-week cohort. Participants co-created a website to educate peers about HPV.

Conclusions: Participants responded to and preferred prevention-focused language. Many participants had no prior HPV-related education and were concerned about associated risks of contracting HPV, with minimal participants expressing vaccine hesitancy specifically. Workshop best practices include providing clear explanations of activities, time for information processing, and encouraging participants to ask questions.

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Introduction

HPV is the most common sexually transmitted infection (STI) in the US; in fact, most people who are sexually active and unvaccinated will get HPV at some point in their lives [1]. HPV education and prevention are vital; incidence of HPV-related cancers increased significantly by an annual percentage change of 0.5% between 2005 and 2019, for all sexes [2]. Within the last three decades, HPV-positive oropharyngeal cancer has increased in incidence by 225% [3]. The most common HPV-associated cancer is cervical cancer, for individuals with a cervix. Cancers of the head, neck, and throat (oropharynx) are the most common HPV-associated cancers for all sexes [2]. In New York, 2,821 people are diagnosed with an HPV-related cancer annually; 59% of cases are in females and 41% are in males [4]. The HPV vaccine is effective at preventing these cancers; people who receive the HPV vaccine between 9 and 12 years old are able to prevent more than 90% of HPV-associated cancers [2]. Globally, more than 270 million doses

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of the vaccine have been administered worldwide since the vaccine was first introduced for the public in 20064.

Most individuals who contract HPV are infected while they are between the ages of 15 and 24 years old [5]. There is a need for targeted youth education surrounding what HPV is, as well as what the HPV vaccine does, especially in terms of the vaccine's protection against associated cancers [6]. Many youth and young adults do not understand the sexual risk factors (i.e. unprotected sex) that can cause HPV infection, nor the associated health risks with HPV2. According to a systematic review of 21 studies published between 2016 and 2019, many youths lack adequate understanding that the HPV vaccine protects against associated cancers [3].

Co-design is an evidence-based approach that has shown promise in its use for developing various educational programs among youth [7,8]. Co-design uses an iterative process to incorporate collective creativity and support the end users utilizing the product of co-design [9]. Co-design approaches are largely informed by two frameworks: social cognitive theory [9] and human centered design [9]. Human centered design aims to create a product ingrained with the requirements and preferences of the stakeholder through an interactive process of iterative engagement until a viable and sustainable intervention is agreed upon [8]. Researchers often collaborate with additional organizations, such as a community organizations, as well as additional stakeholders, such as parents, school personnel, and community leaders, to create inroads to youth members and ensure intervention uptake and widespread implementation [7,8]. This paper provides an overview and quality assessment of a youth co-design health literacy project implemented by the Vaccine Literacy Campaign (VLC) at the City University of New York Graduate School of Public Health and Health Policy. The aims of the project were two-fold: 1) engage teens in co-design activities to co-create resources that promote vaccine literacy, and 2) increase youth confidence and health literacy in relation to HPV and HPV vaccination.

An important aspect of youth co-design is to establish trust and rapport between the participants and the research team. The VLC partnered with the Boys and Girls Club of Harlem (BGCH) to provide HPV education *via* co-design. Members of both organizations met at a youth-oriented community health fair in Harlem, Manhattan, in New York City and began discussing potential collaborations. Months later, the resulting co-design initiative sought to develop youth-driven approaches to vaccine communication and education considering the often-fraught discussions about vaccination in the wake of the COVID-19 vaccine distribution. The BGCH is a 501c3 non-profit organization located near CUNY SPH in Upper Manhattan. BGCH offers after school programs designed to foster both personal and academic development among youth in the local community, ranging in age from 5 to 18 years.

The co-design sessions between the VLC and the BGCH aimed to create a tangible health communication product or campaign with the youth and assess changes in the youths' attitudes and knowledge of HPV and HPV vaccination. This paper will report on the two project aims' outcomes *via* evaluation results and discussion, and description of the final products.

Methods

Implementation

The VLC and BGCH partnership were built over a series of conversations exploring potential collaborations to engage youth in vaccine literacy promotion. The STEM Program manager at BGCH recognized the importance of incorporating health education offerings into their programming and welcomed the VLC's co-design approach. The students at BGCH are unique in that they have agreed to a membership model that holds them accountable for completing the projects they chose to commit to continue to benefit from access to the club and its educational and professional development offerings. At the same time, they have the agency to choose the projects that appeal to them and have blanket parental consent to engage in projects at BCGH. The CUNY School of Public Health Institutional Review Board waived informed consent as the project fell under quality assessment and was not considered human subject's research. This allowed the VLC to feel confident in the students' engagement and focus on creating compelling and creative activities to meet their expectations and honor their commitment. Throughout the

project, the VLC followed the lead of the STEM Program Manager regarding appropriate activities and advice regarding sustaining attention among the teens.

With the help of the STEM Program Manager at BGCH, the VLC facilitators recruited participants through information sessions and flyer distribution to club members. Thirteen students were recruited, aged 16-18 years old. The co-design sessions took place on-site at the BGCH center. The HPV co-design was comprised of 6 sessions, including an introductory session and 5 co-design workshops. Each participant received a $50 gift card at the end of each week following completion of the workshops and a feedback survey. The workshops ran from March 2nd, 2023, to April 6th, 2023. The number of attendees for each workshop averaged 12 students with 13 unique students engaged across the six weeks.

Prior to recruitment and engagement with the BGCH participants, the VLC team completed in a rapid review of the literature pertaining to youth-centered co-design sessions to review best practices. The rapid review included a search of 28 studies published between 2019 to 2023. The most common co-design activities for youth used collaborative small-group work that involved writing responses to prompts [10]. The co-design efforts were conducted with youth and young adults ages 6 to 24 years old while the typical ages were 12-24 years-old. Cohort sizes ranged from four to thirty people [7,8]. Common activities included 'barrier posters', in which participants wrote down answers to a question prompting them to consider barriers towards receiving care. Barrier posters and word-clouds were used to catalyze discussion amongst co-designers to produce solutions while building off each other's ideas. The rapid review of co-design sessions found that many evaluations were primarily completed through analysis of co-design products created during the sessions, (e.g. idea maps, concept walls, jam-boards, concept maps) used to gather qualitative feedback and insights on session feedback [11–15]. In rare cases, evaluation methods included an outside observer able to assist the research team with ensuring key ideas and themes were captured during the qualitative analysis [16].

The study design was reviewed by the CUNY School of Public Health Human Research Protection Program Office and was deemed exempt from IRB review as a Quality Assessment project. During the evaluation, participants were informed that anonymous quotes would be used for product development.

Three tables in the appendix (Online Appendices 1,2 and 3) illustrate the planning documents used to map out the inputs, activities, outputs, and outcomes [8,10,11,12,14,17–20]. The Co-Design Implementation Checklist recounts important considerations for curriculum development and provides a detailed overview on the three phases, pre-co-design, during co-design, and post-co-design. A Co-Design Logic Model demonstrates projected short and long-term outcomes, and the Co-Design Timeline overviews the project roll out. The timeline details the weekly focus of the sessions and the evaluation measures that took place after each week. The participants were engaged in co-creation to begin product development as soon as week two. The facilitators and BGCH provided materials to the students, including paper, pens, markers, and index cards.

Co-design workshops

Over the course of six workshops, the teens discussed their own perceptions and questions about HPV vaccination, their preferred sources and platforms for health information, and what messages and messengers helped them feel confident about vaccination. The introductory session took place prior to the start of the five-workshops and provided an overview of co-design and HPV education on topics including HPV transmission, infection, related cancers, and vaccination. The co-design activity at the end of session one included a rapid barrier-mapping activity in which participants wrote down responses to the prompt '*what are some reasons why someone your age may not be vaccinated for HPV?*' The sticky notes were read aloud to the group and then the teens identified emerging categories of barriers. After, in small groups, they brainstormed possible solutions to tackle each category.

The first workshop began with an educational presentation to address the many questions the teens still had about HPV and HPV vaccination following the introductory session. The co-design activity reviewed the barriers determined the week prior and discussed whether to change, add or remove barriers. Once the group reached consensus on the barriers identified, they were again sorted them into categories (i.e. lack of education). Four categories were agreed upon and in small groups, the teens listed possible solutions that could mitigate each category such as integrating HPV education into health

classes in high school, or social media advertisements. Following the workshop, the teens received an online survey that asked them to reflect further on each barrier category.

Workshop two began with a presentation that summarized the participant solutions from the previous workshop along with those compiled from the reflection survey. In small groups, the teens then discussed various media or materials that could be facilitate the proposed solutions (i.e. social media).

In workshop three, the participants were asked to vote on the materials they wanted to create together, narrowing the list down to three main ideas. In three groups, the teens used prototype planning to describe the user/audience experience, the resource's content and design, and the steps required to create and distribute the resource. These plans were then presented to the full group, followed by a survey to gather feedback on each prototype.

After discussing survey feedback in workshop 4, three key materials were identified by the teens: a poster to be distributed in school and clinic settings, a brochure that could be handed out in high schools, and social media assets. Initial discussions considered presenting all three materials as a package, followed by small group discussions to determine text, graphics, and references for the content. Each group used tablets to search the internet for compelling messages and design inspiration from existing resources. Teens also referenced the educational presentations from previous sessions. Following the workshop, the VLC compiled the teens' ideas and created prototypes of each component using their design and messaging elements.

The teens were presented with the first drafts of the suite of HPV educational materials (brochure, poster, and social media carousel) during workshop five. They reviewed the drafts and provided edits to refine the formatting and messaging content.

The final workshop was used to review the near final versions of the suite of HPV educational materials. Based on feedback from workshop five, the VLC created a simple website to house the educational content, poster and social media assets for downloading and sharing. The brochure then became a simple document, with mostly figures and QR codes to drive teens to the website. The participants provided constructive feedback to improve the website and discussed their plan for distribution and promotion.

Evaluation measures

The evaluation activities were broken into two components, 1) a questionnaire to garner feedback of each co-design session and continue creative design activities, and 2) a survey to monitor changes in confidence and knowledge related to HPV and HPV vaccination. The questionnaires were distributed weekly after each workshop to gauge satisfaction with the workshop content and activities. The confidence and knowledge questions were administered during the first and after the final sessions to compare baseline knowledge with knowledge gained and to understand the impact of the workshop participant on participants' confidence levels with the education.

The confidence questionnaire included seven questions asking participants to rank their confidence on a five-point Likert scale ranging from 1 (least confident), to 5 (most confident). Confidence questions were as follows: 1) How confident are you in your ability to explain what HPV is? 2) How confident are you in your ability to explain the potential long-term impacts of HPV infection? 3) How confidently can you explain how HPV vaccine protects against disease? 4) How confident are you in your ability to explain HPV Vaccination recommendation? 5) How HPV affects female bodied people? 6) How HPV affects male bodied people? And 7) How confident are you in your ability to recognize false or misleading information about vaccination? The confidence questions were to determine the impact the co-design sessions had on the youth's ability to self-report confidence on several HPV-related statements.

Evaluation of co-design studies is primarily performed through qualitative analyses of the outcomes/products (ex: idea map, concept walls, discussion information) produced during co-design sessions [8,11–16]. In addition, co-design evaluation is typically completed through qualitative data analysis of the discussions during the co-design workshops, alongside worksheet responses, where applicable. Sometimes, co-design or participatory methodologies also included an outside observer able to assist the research team with ensuring key ideas and themes were captured during qualitative analysis [11]. Very often, evaluation takes place during the co-design session in which the students would use visual mapping to gather qualitative feedback and insights [12–16]. During the HPV co-design pilot, to keep records of workshop

activities, student participation and design development, the facilitators took careful fieldnotes following each session. All workshop products (i.e. white boards, sticky notes, group drawings) were collected, photographed and documented in a document following each session. Frequently, if the entire participant group was brainstorming or designing together, the facilitators would use a shared document, projected for all to view, to record project development in real time.

Results

The participants of the co-design session succeeded in designing a final product – a website, to educate their peers about HPV. Using the materials provided during co-design, the participants came together to create a list of important educational topics and presentation ideas to educate on HPV. The final content of the webpage included information on what HPV is, how it is transmitted, long-term health risks and outcomes, and information on prevention. The website is available here, and images captured from the webpages are in appendix 4.

Open ended evaluation questions were distributed after each session to prompt reflection and gain qualitative data from the participants. Week two prompted participants to 'Use the space below to describe what you would like to create to help teens feel informed, comfortable, and confident about HPV and HPV vaccination. Think about things that have helped you make a healthy decision in the past – was it a conversation, a video, a poster, a social media post, etc.? Or maybe a combination of a few things? Use these thoughts to help inspire your design'. The responses ($N=5$) referenced the importance of diversity in terms of marketing to different audiences by including different races, ethnicities, and genders, to connect to a wide range of audiences. In the post-session evaluation for week two, one participant wrote in response to the prompt 'What you would like to create in order to help teens feel informed, comfortable, and confident about HPV and HPV vaccination' that 'The problem arises with people being reluctant to learn about HPV, thinking it's a waste of time or it's boring. So, the question that needs to be asked is simple; how do we get people to WANT to learn about HPV and how to prevent it (not to mention the vaccine too)? For the reflection prompt 'Describe what would like to create to help teens feel informed, comfortable, and confident about HPV and HPV vaccination. Think about things that have helped you make a healthy decision in the past – was it a conversation, a video, a poster, a social media post, etc.? Or maybe a combination of a few things? Use these thoughts to help inspire your design' 83% of respondents [$N=6$] referenced utilization of social media to spread awareness and education.

Additionally, when prompted for feedback on how to best market the education, a recurring theme amongst participant responses included incorporating a medical professional, particularly a doctor, to validate the information. These mimicked results seen in other youth co-design studies, which emphasize the importance of education from both healthcare professionals, as well as individuals diagnosed with the health condition [21,22]. Through facilitated dialogue at the co-design sessions, as well as in the qualitative evaluations, participants expressed desire to hear from people with lived experience of having HPV and/or having the HPV vaccination. At the end of week five, a post-evaluation was distributed, with a question stating 'Please describe whether you feel like your voice is being heard during the session discussions. Do you think that your opinions are being considered and valued?' All participants ($N=9$), completed the open-ended questionnaire and stated they felt their voice was heard and valued, 'I feel like I contributed a lot'. One participant said, 'I am satisfied with how I was asked to participate, and I wanted to participate because I felt I would have an impact'.

The first session asked participants ($N=13$) to write down responses to the facilitator lead prompt 'reasons a teenager may not get the HPV vaccine'. Responses were grouped into four major themes, 1) Lack of HPV Education or Miseducation about vaccines, 2) Missing, confusing or misleading sources of information, 3) Doubt in the efficacy or necessity of the HPV vaccine, 4) Fear - of needles, vaccine safety, or of the discussing the topic with others. Images of student co-design work are in appendix 5, Co-design Activity Outputs. Overall, there was an increase in confidence in explaining HPV-related information to others by the youth co-design cohort, see Table 1. Participants received the confidence evaluation after the first session and after the fifth session. The evaluation response options were a five-point Likert scale ranging from 1 – not at all confident, to 5 – extremely confident. The percentage increase ranged from 18% to 72% increase. The biggest increase in confidence was in the ability to explain how HPV impacts women. This may be related to the fact that the co-design participants were 85% male, explaining the lower confidence originally in explaining HPV vaccine information related to female bodies.**Co-design**

Table 1. Confidence evaluation.

Question	Week 1 Mean (SD)	Week 5 Mean (SD)	
Score	*1 = Not at all confident*	*5 = Extremely confident*	Percent Increase
What human papillomavirus (HPV) is and how it is spread	3.78 (0.70)	4.11 (0.57)	9.1%
The potential outcomes of HPV infection including possible cancers	3.92 (0.73)	4.11 (0.74)	4.8%
How the HPV vaccine protects against infection	3.92 (0.62)	4.33 (0.67)	10.5%
What the current HPV recommendations are for different ages	3.92 (0.83)	4.00 (1.05)	2.0%
How HPV can impact girls/women	3.46 (0.93)	4.11 (0.74)	18.8%
How HPV can impact boys/men	3.85 (0.95)	4.11 (0.87)	6.9%
How confident are you in your ability to recognize false or misleading information about vaccination?	3.15 (1.29)	3.89 (1.19)	23.3%

Co-design Session: Satisfaction Results

Question	Overall Mean (SD)	Variance
Group activities	4.43 (0.83)	0.33
Facilitator's presentation	4.64 (0.76)	0.33
Workshop goals	4.12 (1.15)	0.00
Expectations for next session	4.17 (1.07)	0.92
Educational information	4.38 (0.91)	0.00

Youth confidence in HPV associated knowledge increased over the five-week codesign cohort. The confidence questions, measured on a five-point Likert scale, were self-reported measures by the participants, taken in an evaluation distributed after the first and last session. The average confidence across all seven measures, outlined below, ranged from 1 (not at all confident) to 5 (extremely confident). The confidence on all measures increased in ranges from 2% to 23%. The question 'What the current HPV recommendations are for different ages' increased only by 2%, however, participants self-reported high confidence on this measure (week 1 = 3.92, week 5 = 4.00), with the max score being 5.

Discussion

The study was conducted with the goal of creating co-design materials to appeal to youth, who are the most in tune with approaches that would best work amongst this demographic. Strengths of the co-design implementation included the strong collaboration amongst participants and between participants and facilitators. The iterative process of the co-design session led to a final product development that participants evaluated positively. The evaluation results demonstrated participants felt engaged and as if their voice mattered, facilitating an environment in which participants were empowered to speak their mind, provide disagreements on productive development, to yield a product everyone liked. It was clear early on that the participants would benefit from additional educational content on HPV and the vaccine, and the facilitators were able to pivot rapidly to meet this need. Additionally, given that this cohort was familiar with each other through pre-existing relationships developed from attending the Boys and Girls Club, there was an existing rapport between co-design contributors. This mitigated the need to acclimate participants with each other, fostering effective collaboration from the start. Participants were very comfortable providing feedback on the product, presumably due to existing relationships. Limitations of the study mainly relate to the quick turnaround time from session one to product development. Time was limited due to the need for background HPV education and sessions not starting on time.

With some lessons learned, the project team offers recommendations for future iterations of co-design sessions with youth to improve participant focus during sessions and increasing survey completion rates. We recommend a baseline knowledge assessment prior to any co-design activities. This will help determine whether and to what extend educational content needs to be included and tailored to participant needs. Educational sessions may be needed to ensure all participants have a similar understanding of the health issue. Momentum going into the HPV co-design workshops was hindered by the lack of HPV knowledge among the participants. The brief educational overview allotted in the timeline was not sufficient and led to additional time spent on education in the second session. We recommend comprehensive overview (if the need is determined by the assessments) during the first session, as well as providing pre-reads, brochures, and informational webpages for youth on HPV content and education to supplement the education and mitigate time allocated on questions. Providing education prior to initiation of co-design has been a positive asset in prior youth co-design studies [6]. Additionally, it is important to

leave ample time for completion of co-design activities during workshops and allow time for clarification of instructions. If possible, having a flexible timeline is also beneficial and allows for the inevitable adjustments need to ensure all participant feel comfortable with the health topic and have sufficient time to explore their ideas. Recommendations are consistent with the broader literature surrounding youth co-design studies. There is an emphasis on the inclusion of the target population of health-intervention in the codesign process [22]. While literature on the implementation of youth co-design best practices is scarce, many health-related interventions focus on mental health and risk reduction, inclusive of sexual health [22]. In terms of specifically digital health-related youth co-design interventions, the development of a website is most common [22]. Additionally, there is a need for evaluative information regarding co-design participant experience, which is also lacking in the literature [22]. This study reports on the participant experience throughout the entire course of co-design, which was utilized to aid the development process of the final HPV-education product. Additionally, we recommend partnership with a local community-based organization to aid in the development of establishing rapport against the co-design cohort. Prior youth co-design interventions have allocated time towards team bonding [13,22], however, the need for this was mitigated by partnership with a pre-establishing community organization, Harlem Boys and Girls Club, which allowed for more time to be devoted towards the implementation of the co-design.

A final recommendation concerns incentives. The gift card incentives required that weekly compensation to be delivered in one payment. However, it would have perhaps been more effect if the gift card payments were structured to provide half the payment for attending the co-design session itself and half the payment for completing the follow up and feedback activities after each session. While the evaluation response rates were overall high ($M = 81\%$, SD = 18%), this required an administrative burden on the co-design facilitators to remind the students to complete the evaluations.

Overall, the co-design workshops were effective in meeting the stated objectives: 1) engage teens in co-design activities to co-produce a health literacy resource and 2) increase youth confidence and health literacy in relation to HPV. On average, the students' confidence in their ability to explain what HPV is increased across *all* six measures on the pre versus post confidence scale. Additionally, based off participant qualitative feedback, social media is an effective outlet for teen and youth communication for health literacy. Two key take-aways are that 1) co-design is an effective strategy to educate and engage youth on the HPV vaccine, and 2) co-design can be used by youth to create an educational product that targets the participant demographic.

Authors' contributions

JD - Contribution to conception and design of co-design, writing manuscript content, evaluation analysis. LR - Major contribution to conception and design of co-design, critical revisions, manuscript content, evaluation analysis. KN - Conception and design of co-design, critical revisions, evaluation analysis. All authors agree to be accountable for all aspects of the work.

Disclosure statement

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Data availability statement

The data is available upon a reasonable request from the corresponding author.

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Appendix A

Table A1. Co-design activities.

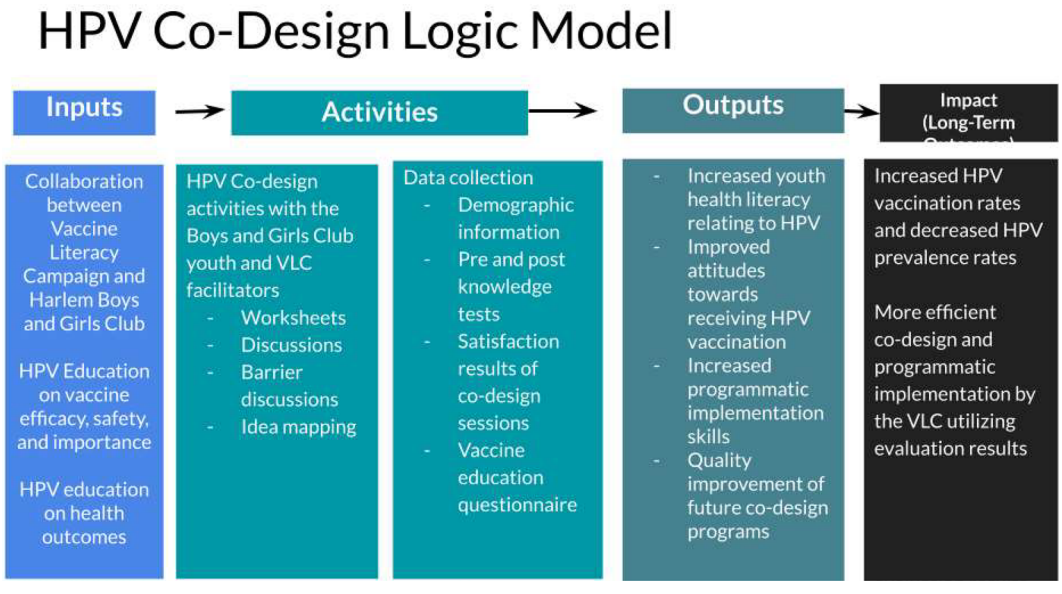
Activity name	Description
Barrier posters	Participants are asked to write down responses to a prompt regarding barriers to care and place the written responses on a white board at the front of the room. This opens the door for discussion on how to overcome barriers and move towards improving healthcare[17].
Word clouds	Youth are use technology to answer questions on and the answers to the questions were presented on a word cloud in the front of the classroom[18].
Incorporation of mainstream media	Researchers and facilitators incorporated mainstream media ideas that are relevant to youth into the codesign process[14].
Cartoon Captioning	Cartoon caption was an activity used with high-schoolers to better understand the types of messages that adolescents receive from adults regarding decreasing risk for diabetes and explore how they feel about these messages[17].
Building rapport	The purpose of this aspect of the co-design session was to build rapport and establish trust among the co-design group, can be completed in the form of time allocated to eat and socialize, typically with no formal agenda[14].
Cool wall	Cool walls are used for participants to work together to rank an item/topic, etc., from "cool" to "uncool"[14].
Worksheets	Worksheets of phone screen templates, magnetic whiteboards, and magnets to write, sketch and/or paste in user interface components[12,15,19].
Sorting and ranking activities	Co-design participants receive physical items with writing on them and are asked to rank in order of preference[8,20]. Sorting and ranking activities help uncover why participants trust certain health-related situations over others and identify barriers and facilitators for intervention adherence[8]. Ranking and sorting can be used in lieu of typical Likert-scale reported responses [8].
Concept mapping	Create a concept map [write down ideas and place them on a wall} followed by group discussion[10,12,14].
Informal discussion	Facilitators will initiate a discussion with prompts for participants to lead a discussion based off of their feedback[18].

Appendix B

Table B1. Co-design implementation checklist.

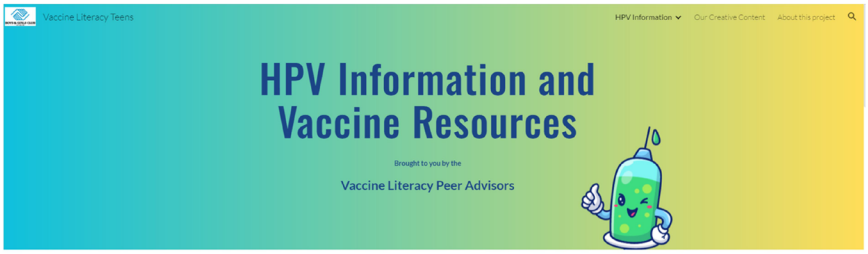
Pre-Co-Design Sessions	• Gather information on current trends on youth co-design • Presentations and information packaged to present to youth relating to HPV, HPV vaccine • Gather physical supplies necessary for co-design (ex; markers, pens, paper, etc.,) • Draft curriculum for co-design that can be flexible based off student needs • Create and administer a baseline knowledge assessment
During Co-Design	• Provide a session plan to all of the co-design team and clearly explain expectations and outcomes • Tailor educational component for session one based off of knowledge assessment results • Administer evaluation to co-design team after each session and send a reminder 2 days later to complete the evaluation • Take field notes after each co-design session, that include attendance and pictures of material developed
Post Co-Design	• Evaluation analysis • Share final products and prototypes with the co-design team

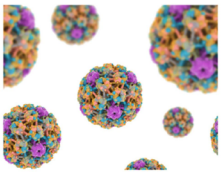
Appendix C



Appendix D. Co-design webpage

https://sites.google.com/view/vaccineliteracyteens/hpv-information





85% OF PEOPLE WILL GET HPV IN THEIR LIFETIME

But what is it? Human papillomavirus (HPV) is a group of more than 200 related viruses. It is highly infectious and can cause cell mutations **that lead to various cancers**. In most cases, this virus presents as genital warts.

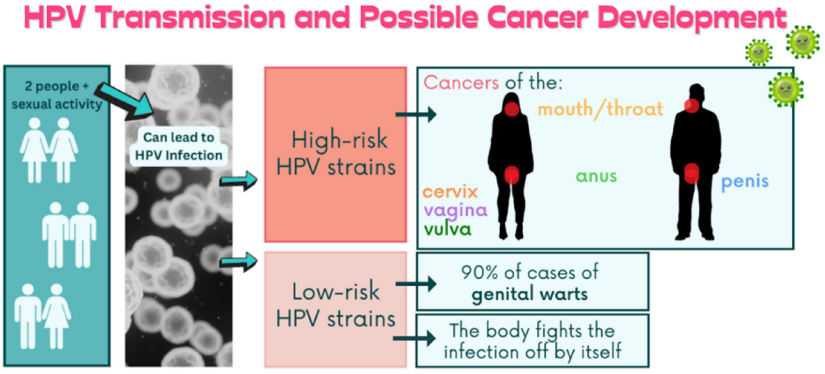
HPV infections are very common and young adults are at highest risk for exposure. **75% of HPV infections occur in people 15-24 years of age.** Scroll down to learn more about HPV and the vaccine to prevent HPV-related cancers.

HOW CAN YOU GET HPV?

HPV is spread through genital or skin-to-skin content, usually through sexual activity. The virus can spread even when no symptoms are present! While most cases of HPV will resolve on their own in under two years, some **high-risk strains can lead to cancer.**

There are six types of HPV-related cancers: cervical, vaginal, vulva, penile, anal, and oral. **HPV causes more than 36,000 cases of cancer a year In the United States.**

HPV Transmission and Possible Cancer Development



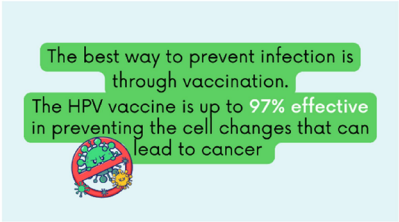


HPV INFECTIONS AND DETECTION

There is no treatment for HPV infection but there are procedures to remove damaged cells if the disease progresses. **Genital warts is how most people find out they have HPV.** Genital warts look like red bumps on your genitals, they can vary in size.

Usually HPV is detected for females via a pap smear at the gynecologist. The pap smear is a cervical cancer screening. Anal pap smears are also given to populations who have a higher risk of anal HPV infection. Cancer is commonly screened for at other routine physician visits, including dental exams, who will exam your mouth and throat.

Fortunately, HPV vaccination is highly effective in preventing infection and preventing the development of related cancers.



Appendix E. Co-design activity outputs

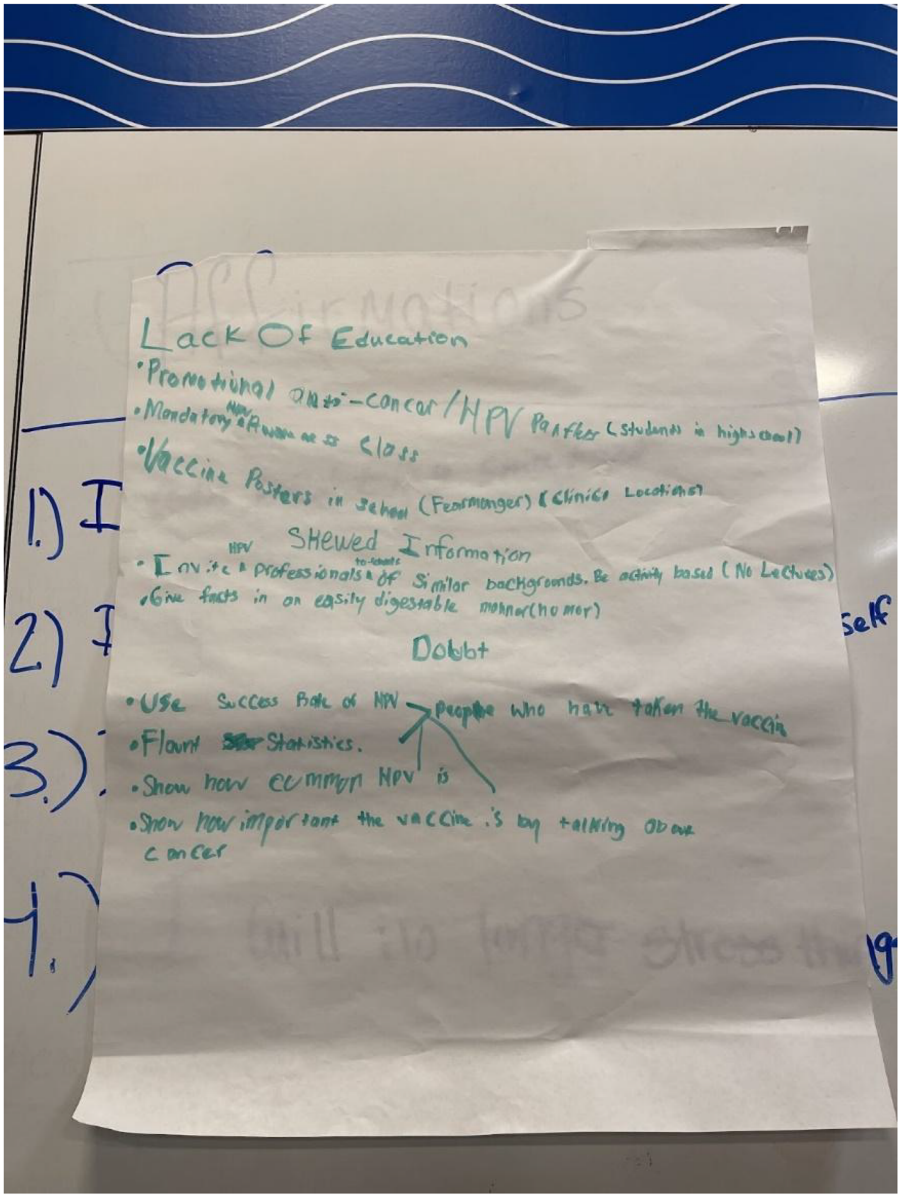


Image taken from week three one session with Harlem Boys and Girls Club.

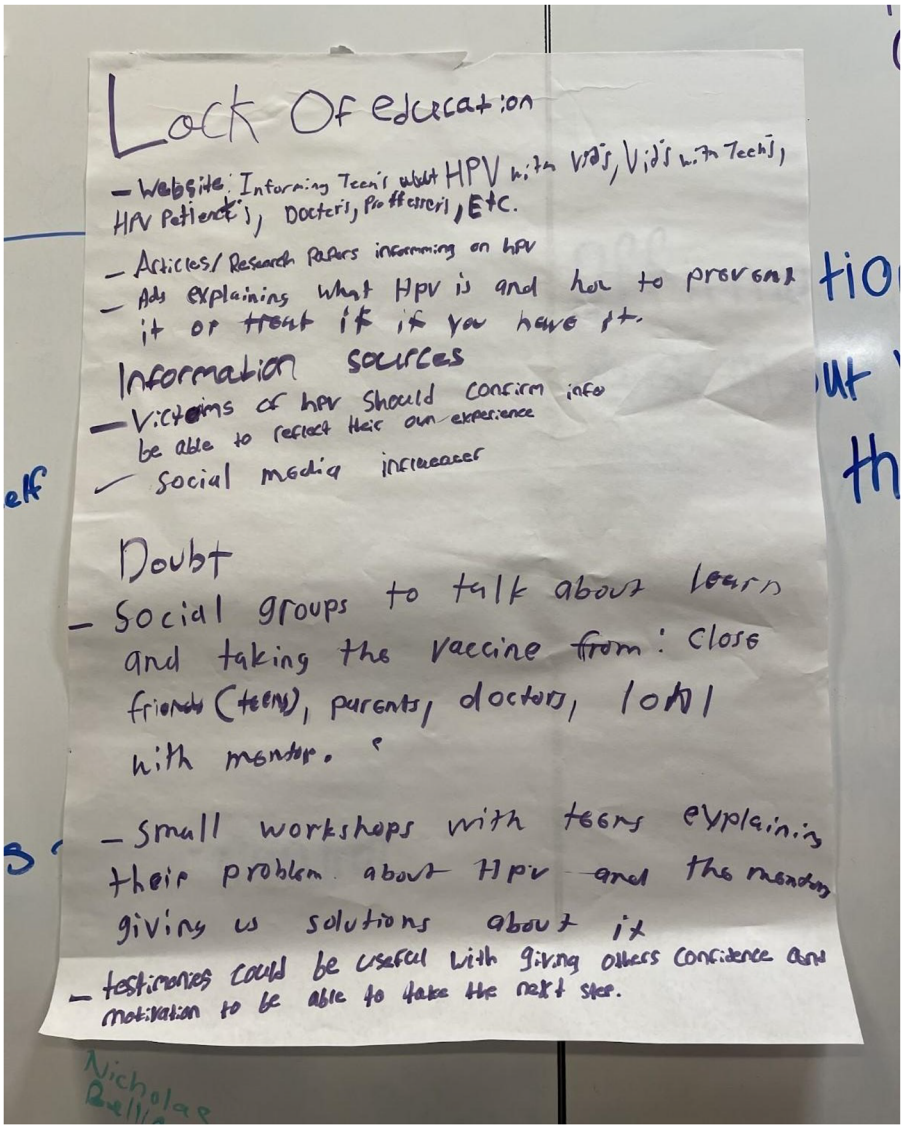


Image taken from week two co-design session with Harlem Boys and Girls Club.

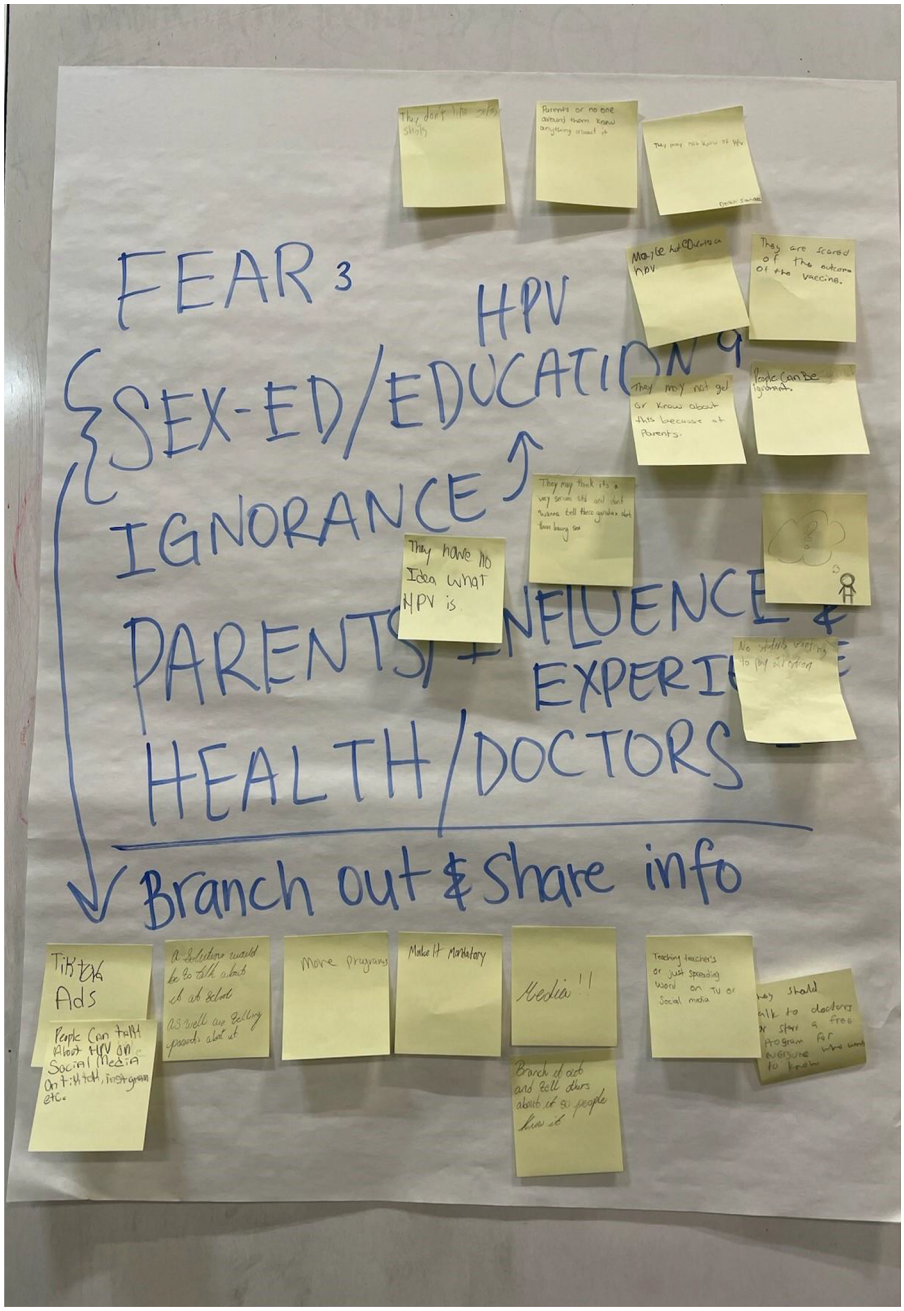


Image taken from week three co-design session with Harlem Boys and Girls Club.